TEXORA · LECTURE SHEET

Concept of IE: Industrial Engineering in the Apparel Industry

Work study applied to garment making: method study and work measurement, the construction of the standard minute value from observed time, rating and allowances, and how IE data drives capacity, costing, balancing and incentives.

AM-4201 · ADVANCED APPAREL MANUFACTURING-II · TOPIC 3 · LEVEL 4 TERM 2

1. What industrial engineering is

Industrial engineering (IE) is the discipline of designing and improving integrated systems of people, machines, materials and information so that work is done by the best known method, in a measured standard time, at the lowest sustainable cost. In apparel — where direct labour is the largest conversion cost and the product changes weekly — the IE department is the factory's measurement authority: it owns the **SMV database**, sets line targets, balances lines, validates CM costings and leads method improvement. Every number in Topics 1, 4 and 6 of this course is an IE output.

IE's classic instrument is **work study**, two techniques used in order: **method study** (find the best way) then **work measurement** (time the best way fairly). Timing a bad method merely standardises waste.

2. Method study in the sewing room

Method study follows the SREDIM sequence (Select, Record, Examine, Develop, Install, Maintain): select an operation (high SMV, bottleneck, quality-critical); record it with charts and video; examine with the 5W1H questions and motion economy principles; develop the improved method; install with training; maintain by audit. In apparel its typical wins are unglamorous and large: re-positioning bundle and trolley to kill reach motions; templates, guides, folders and attachments that move skill from the hand into the jig; combining or eliminating operations (a fused interlining removes a tacking operation); workstation layout so the garment flows without re-orientation. A 3-second saving on an operation repeated 500 times a day is 25 minutes of capacity — method study is arithmetic with a stopwatch and a conscience.

3. Work measurement: building the SMV

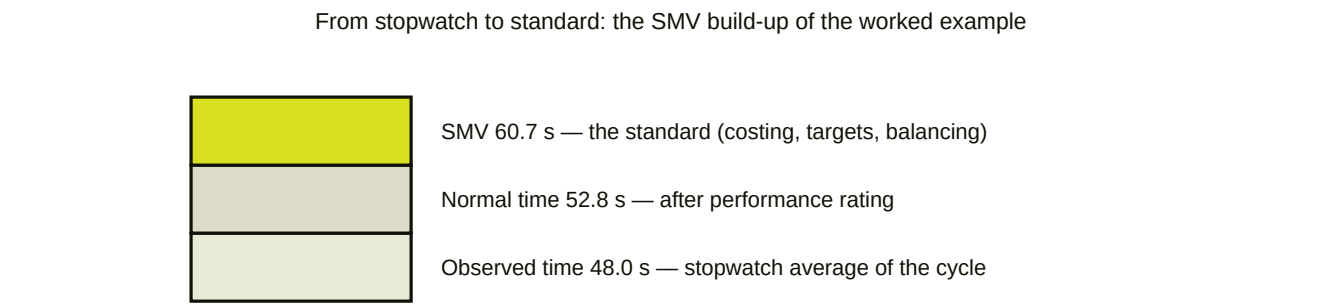


Figure 1 — The three layers of a standard minute value. Skipping the rating layer is the commonest cause of "impossible" targets.

WORKED SMV — VERIFY EVERY NUMBER

Operation: attach pocket (cycle studied over 20 cycles). **Observed time** (average) = 48.0 s. The operator is rated at **110%** of standard pace (working faster than standard): **Normal time** = observed × rating = 48.0 × 1.10 = **52.8 s**. Allowances: relaxation (personal + fatigue) and contingency total **15%**: **SMV** = 52.8 × 1.15 = **60.72 s = 1.012 min**. Garment SMV is the sum of its operation SMVs from the operation breakdown; line targets, costing and balancing (Topics 1 and 4) all divide by this figure, so a 5% error in rating propagates into every plan the factory makes.

Two measurement routes exist. **Stopwatch time study** measures real operators (rules: enough cycles for statistical confidence, rate every cycle, discard abnormal readings with recorded reasons). **Predetermined motion time systems** — in apparel, **GSD (General Sewing Data)** and MTM-based systems — build times from coded standard motions without a stopwatch: faster for new styles (times exist before the first garment is sewn, from the operation description), consistent across factories, and the normal basis of pre-production costing. Mature IE departments cross-check both.

4. Where IE data goes

APPLICATION	HOW THE SMV IS USED
Capacity & target	Hourly target per operator = 60 ÷ operation SMV (at 100%); line target from the master equation of Topic 1 at the planned efficiency.
Line balancing	Operations assigned so each station's total SMV ≈ pitch time = garment SMV ÷ operators; imbalance is visible as station idle time (Topic 4).
Costing	CM price check: garment SMV × factory cost per minute must fit inside the buyer's CM (Topic 1).

Incentives	Piece rates and bonuses are earned minutes ÷ attended minutes — impossible without trusted SMVs, and explosive with wrong ones.
Method improvement	Before/after SMV is the proof a method change worked; savings are banked in the database, not in anecdotes.

5. The human contract of IE

Standards govern people's earnings and effort, so IE is practised under rules: studies are announced, not secret; rating is a trained, calibrated skill (IE staff rate standard films to stay honest); allowances follow policy, not negotiation per operation; and SMVs change when the *method* changes, never merely because output is wanted higher. A factory whose operators distrust the stopwatch will defeat every plan built on it — IE's authority is technical and ethical at once.

Glossary

TERM	MEANING
Observed time	Average measured cycle time of an operation as performed.
Rating	Assessed pace of the observed operator as % of standard performance (100%).
Normal time	Observed time × rating ÷ 100 — time at standard pace, without allowances.
Allowances	Added % for rest/personal needs, fatigue and contingencies; SMV = normal time × (1 + allowances).
GSD	General Sewing Data: predetermined motion-time system for sewing operations.
Pitch time	Garment SMV ÷ number of operators — the target cycle time per station in a balanced line.

Key points and summary

- IE = method study first, then work measurement; its database of SMVs runs the factory's planning, costing and pay.
- SMV construction: observed time → (× rating) normal time → (× 1 + allowances) standard time.
- Worked standard: 48.0 s at 110% rating + 15% allowances = 60.72 s = 1.012 min.
- GSD/PMTS gives times before production; stopwatch studies verify them on the floor.

Practice

Answer in your own words; brief model answers follow.

1. An operation averages 36.0 s observed at a rating of 90%. Allowances are 12%. Calculate normal time and SMV (in seconds and minutes).
2. An operator is observed sewing a cycle in 40 s and rated 80%. A trainee takes 50 s at rating 100%. Whose observation gives the larger normal time, and what does that teach about choosing study subjects?
3. A garment has operation SMVs summing to 18.6 min. A method change removes an operation of 0.9 min and shortens another from 1.4 to 1.1 min. What is the new garment SMV, and what happens to the line's daily target at unchanged efficiency?
4. Why must method study precede time study?
5. Give two abuses that destroy trust in an SMV system.

Answer key

1. Normal time = 36.0 × 0.90 = 32.4 s. SMV = 32.4 × 1.12 = 36.29 s = 0.605 min.
2. First operator: 40 × 0.80 = 32 s normal. Trainee: 50 × 1.00 = 50 s. The trainee's is larger — because at rating 100% the observed slowness is real method/skill, not pace. Studies should use trained operators working the standard method at a steady pace; timing learners builds learner standards into the database.
3. New SMV = 18.6 − 0.9 − 0.3 = 17.4 min. Target scales by 18.6 ÷ 17.4 = 1.069 — about 6.9% more garments per day at the same efficiency and crew.
4. Time study freezes whatever method exists into the standard. Timing first locks in wasted motions, poor layout and unnecessary operations — the factory then pays incentive on waste and resists changing a method whose time is already "standard".
5. Examples: cutting SMVs to raise output without a method change; secret studies; rating by favour or pressure; refusing to re-time after a genuine method improvement. Each teaches operators that the numbers are a management weapon, after which data collection itself is sabotaged.

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